

Ara Esfarjani

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EDUCATION

University of Southern California

Los Angeles, CA

B.S. in Computer Engineering and Computer Science

May 2027, GPA: 3.8

- **Relevant Coursework:** Machine Learning, Professional C++, Data Structures, Embedded Systems, Probability/Statistics, Parallel/Distributed Computation, Distributed Systems, Networking, Operating Systems
- **Activities & Societies:** Head of Developer Curriculum, QuantSC (USC's Quantitative Finance Club)

EXPERIENCE

Susquehanna International Group

Bala Cynwyd, PA

Incoming Quantitative Strategy Developer Intern

May 2027 – Aug 2027

- Signal detection, quantitative strategy analysis, and low-latency C++ strategy development on a trading desk.

Axiom Math

Palo Alto, CA

Research Engineer Intern

Aug 2026 – Dec 2026

- Optimize the Lean compiler and elaborator to cut proof-search latency in RL training rollouts.

Coinbase

San Francisco, CA

Software Engineer Intern, Transaction Intelligence

May 2026 – Aug 2026

- Built a Go/gRPC platform on Kubernetes verifying requests flow correctly through downstream microservices.
- Architected a registry-based orchestrator composing cross-team pipelines from pluggable gRPC components.
- Onboarded **7** microservices (Money Movement, Ledger) with thread-safe execution, idempotency keys, auth proxy.
- Persisted runs in MongoDB; debugged failures via Datadog traces; validated in Docker before prod deploys.

Stanford Artificial Intelligence Laboratory

Stanford, CA

Student Researcher, Arbabian Lab

March 2026 – Present

- Automated optimization of TIA receivers by co-designing a multi-objective genetic search (NSGA-II/III) over **30+** circuit parameters with a Cadence circuit simulation backend.
- Scaled evaluation to **80** parallel CPUs, achieving near-full utilization across thousands of simulations per run.

Celestial AI

Santa Clara, CA

Machine Learning Engineer Intern

May 2025 – Aug 2025

- Wrote a C++ profiling harness benchmarking quantized LLM inference over a photonic-fabric interconnect, characterizing latency/bandwidth and accuracy-throughput tradeoffs.
- Deployed vLLM on GPUs, achieving **4×** inference speedup and cutting query latency **60%** via model selection.
- Built a RAG pipeline for NDA clause extraction, cutting legal review time **80%** across **250+** documents.

Pivotal.aero

Palo Alto, CA

Embedded Software Engineer Intern

Jun 2024 – Nov 2024

- Built a blackbox in C/C++20 on flash memory via queued SPI, increasing data throughput **43%**.
- Reduced real-time latency with priority-preemptive tasking and optimized scheduling on FreeRTOS.
- Validated **50+** boards for TCP/IP and CRC32 across UART, CAN, and RS-485 using JTAG and a logic analyzer.

PROJECTS

VRIQ: Benchmarking Visual-Reasoning IQ of VLMs | *Python, PyTorch*

- Co-authored an arXiv preprint at USC's ANRG, contributing **35%** of a synthetic dataset of chart, diagram, and document tasks that break vision-language models.

TECHNICAL SKILLS

Languages: Python, C++ (20/23), C, CUDA, Go, Java, TypeScript, SQL, Bash

Technologies: PyTorch, NumPy, Docker, Kubernetes, Kafka, gRPC, MongoDB, AWS, PostgreSQL, Flask, React

Concepts: AI Agents, GPU Programming, LLM Inference, RAG, Linux